

PROJECT INITIALIZATION AND PLANNING PHASE

Date	04 July 2026
Team ID	SWTID-2026-4860
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal report aims to transform personal and classroom learning using lightweight Generative AI and cloud service orchestration, boosting academic efficiency, self-assessment abilities, and customized learning guidance. It tackles student limitations in accessing heavy hardware or premium models, promising fast offline/local operations and tailored educational support.

Project Overview	
Objective	The primary objective is to revolutionize the study workflow by implementing a fast, efficient AI-powered educational assistant using FastAPI and state-of-the-art Large Language Models (LLMs) to ensure direct concept explanation, question answering, and automated smart-quiz evaluation.
Scope	The project comprehensively builds an interactive interface utilizing local models (LaMini-Flan-T5) and cloud APIs (Google Gemini 1.5 Pro) to handle instant Q&A, comprehensive document summaries, structured roadmaps for technical subjects, and adaptive quizzes.
Problem Statement	
Description	Students face information overload, static curricula, and lack affordable, round-the-clock tutoring support. Existing powerful LLM systems are locked behind costly subscriptions or require continuous high-speed data networks and expensive, power-heavy computational hardware.
Impact	Solving these issues will democratize customized education, allowing any student to get immediate clarity on complex fields (like Math or SQL), reduce exam anxiety through automated testing, and run code efficiently even on low-tier or consumer machines like base Mac M1.
Proposed Solution	
Approach	Employing a hybrid architecture with FastAPI for high-performance API endpoints, combined with an interactive front-end (HTML/CSS/Jinja2). Cloud Gemini is called for complex structured plans, while optimized LaMini runs lightweight tasks efficiently.
Key Features	- Intelligent Question Answering (e.g., immediate geographical or science responses). - Automated interactive Quiz Generation for immediate performance grading.

RESOURCE REQUIREMENTS

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Standard local consumer CPU/GPU processing unit optimized to load lightweight AI pipelines efficiently.	Intel i3/i5 or higher / Apple Silicon (Mac M1 optimized)
Memory	Minimum system operational workspace RAM required to maintain backend serving and local model weights concurrently.	4 GB Minimum (8 GB Recommended)
Storage	Free disk space to save local model configurations, Python library packages, database structures, and runtime caches.	10 GB Free Disk Space
Software		
Frameworks	High-performance, modern web application engine for managing server logic, handling requests, and serving dynamic routing paths.	FastAPI, Uvicorn
Libraries	Machine learning execution platforms, model loader frameworks, and official cloud API development kits.	Google Generative AI SDK, Transformers, Torch, Python 3.10+
Development Environment	Code authoring editor tools, version controlling systems, and command terminals for local running.	Visual Studio Code / PyCharm, Git & GitHub
Data		
Data Source	Knowledge structures, text query inputs, schema layouts, and prompt context configuration structures.	Google Gemini 1.5 Pro API & LaMini-Flan-T5 dataset context